

Phys 20AL Week 5: Waves on a string

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1 Aim for the Experiment

The goal is to figure out the relationship between the velocity of wave propagation, and its tension force. It is also aimed to figure out the coefficient μ - the mass per unit length of the string - based on the measured data.

2 Experimental Setup

2.1 Experiment with Manual Setup

The equipments include: Tape, Long string, meter stick, mass, mass scale, pulley, phone (with slow-motion camera).

The setup process is as follow:

1. Fix the pulley at the edge of the table.
2. Tape one end of the string on the wall, and put the other end on the pulley.
3. Hang suitable mass onto the string at the end of the pulley.

2.2 Experiment with Function Generator

The equipments include: String, meter stick, masses, mass scale, pulley, function generator, string vibrator.

The setup process is as follow:

1. Plugin the function generator's power chord, and connect the string vibrator to the function generator.
2. Fix both the pulley and the string vibrator at the edge of the table, with suitable distance in between.
3. Tie one end of the string on the string vibrator, and put the other end on the pulley.

3 Measurement and Methods of Measure

The length of the string will be measured using meter stick, the masses about to hang on the string (to generate steady tension force on the string). About the wave velocity, there are two types:

- For the one with manual measurement, the time for the wave propagation is measured using the camera recording (for more accuracy it's intended to use slow-motion mode in the camera).
- For the one with function generator, instead of measuring the time for the wave propagation, it's measuring the frequency of the string oscillation generated by the Function Generator.

Thought: The use of the pulley is to direct the string downward, so it directly transfers the force of gravity on the masses into the tension force of the string.

For each of the case, the masses measured and used are as below:

1. For experiment that measures everything manually, the mass is 0.025 kg.
2. For experiment that measures the first resonance frequency with varying masses, the masses used include: 0.055, 0.105, 0.160, 0.205, 0.255, 0.306 kg.
3. For the experiment that measures multiple resonance frequencies of the same masses, the masses used include: 0.100, 0.200 kg.

For each of the case, the string length used are as below:

- For experiment that measures everything manually, the string length is 6.000 m.
- For both experiments with function generator, the string length is 0.865 m.

For the first experiment that measures everything manually, we'll do a total of 3 trials to measure the velocity of the wave propagation.

For the second experiment using function generator and measuring first resonance for varying masses, each chosen mass will also do a total of 3 trials to measure the first resonance frequency. (the idea is to get more experimental data of the overall first resonance frequency).

For the third experiment however, instead of measuring with multiple trials, for each mass we'll perform 1 measurement at each resonance frequency, until it is no longer observable. (This typically happens around the n^{th} resonance frequency, where $n = 6, 7$).

4 Experimental Procedure

4.1 Experiment with Manual Setup

The goal for this section is to get a rough / general observation about the wave propagation, instead of performing a precise measurement.

We'll follow the following procedure:

1. Measure the mass used for tension, and hang the mass at the end of the string that's at the pulley's side.
2. Gently tap the string on the other end (that is taped), but with large enough amplitude that can be observed and recorded by the camera.
3. For the whole process in Step 2, record using the slow-motion camera till the time that the wave has traveled back to the initial starting point.
4. Repeat step 2 to 3 for the total of 3 times.

4.2 Experiment with Function Generator

4.2.1 Experiment Measuring 1st Resonance Frequency

The goal for this section is to derive the velocity of the wave using the resonance frequency, and further derive the string's mass per unit length constant μ using the known velocity and tension.

We'll follow the following procedure:

1. Fix a mass used for tension, measure the mass, and hang it at the end of the string (on the pulley's side).
2. Start the function generator, and let the string vibrator vibrate the string.
3. Tweak the output frequency on the function generator, until the first resonance is observed (which requires the string to have stable amplitude, and the pulley is not oscillating back and forth; or more precisely, when the string has a fixed point at the end). Record this first resonance frequency displayed on the function generator.
4. Repeat Step 2 to 3 for a total of 3 trials for the fixed mass.
5. Change the mass mentioned in Step 1, and repeat Step 2 to 4 for each desired mass.

4.2.2 Experiment Measuring n th Resonance Frequency

The goal for this section is to observe multiple resonance frequency when fixing a specific tension force, observe the change of wavelength under different frequencies, and also see if the wave velocity is independent from the resonance frequency (when fixing tension).

We'll follow the following procedure:

1. Fix a mass used for tension, measure the mass, and hang it at the end of the string (on the pulley's side).
2. Start the function generator, and let the string vibrator vibrate the string.
3. Tweak the output frequency on the function generator, until the first resonance is observed (which requires the string to have stable amplitude, and the pulley is not oscillating back and forth; or more precisely, when the string has a fixed point at the end). Record this first resonance frequency displayed on the function generator.
4. Repeat Step 3 inductively for each n^{th} resonance frequency (for $n = 2, 3, \dots$) until the resonance is no longer observable.
5. Change the masses to other ones await to be measured, and repeat Step 2 to 4 for the desired mass.

Length (m)	7.000		length used (m)	6.000			
String Mass (g)	3.200						
Mass for Tension (g)	25.000						
velocity							
total mass (g)	28.500						
Tension (N)	0.279						
mu	0.005						
calculated velocity (m/s)	7.671						
Trial	start (s)	end (s)	t_diff (s)	v_measured (m/s)	uncertainty in time	uncertainty in v	
1.000	4.760	5.420	0.660	18.182	0.020	4.009	
2.000	11.560	12.500	0.940	12.766	0.020	2.815	
3.000	7.820	8.370	0.550	21.818	0.020	4.810	
Length (m)	0.865						record: Hz when in first resonance
Trial \ Mass (kg)	0.055	0.105	0.160	0.205	0.255	0.306	
1	28.710	33.360	40.190	46.730	51.710	57.480	
2	28.520	33.320	40.230	46.660	51.640	57.440	
3	28.560	33.370	40.220	46.640	51.720	57.460	

5 Experimental Data

5.1 Experiment with Manual Setup

5.2 Experiment with Function Generator

5.2.1 Experiment Measuring 1st Resonance Frequency

5.2.2 Experiment Measuring nth Resonance Frequency

Length (m):	0.865						
Fixed Mass 1 (kg):	0.100		Tension 1 (N):	0.981			
Resonance (n)	1	2	3	4	5	6	7
Frequency (Hz)	32.260	64.150	96.380	128.700	161.600	195.900	229.700
Fixed Mass 2 (kg):	0.200		Tension 2 (N):	1.961			
Resonance (n)	1	2	3	4	5	6	
Frequency (Hz)	46.030	90.450	135.400	182.900	228.500	275.000	